

CRIME TREND ANALYSIS IN LOS ANGELES 2020-2024

Data Visualization & Analytics Capstone Project

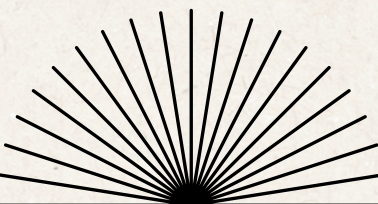
Dataset: LA City Crime Data (2020–2024)

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Group 13 – Section E

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Context & Decision Problem

CONTEXT

- Crime trends in Los Angeles vary across years, locations, and time periods, making it difficult to plan effective public safety strategies without data-driven analysis.
- Understanding hotspot areas and peak crime patterns is essential for improving policing efficiency and citizen safety.

BUSINESS PROBLEM

- Los Angeles faces uneven crime distribution where certain areas and crime types contribute disproportionately to total incidents.
- Without structured analysis, law enforcement cannot effectively prioritize patrol deployment, prevention initiatives, and investigation resources.

DECISION OBJECTIVE

- The objective of this project is to identify year-wise crime trends, dominant crime categories, hotspot areas, peak time windows, and case closure performance using crime data from 2020–2024.
- These insights will support targeted decision-making to reduce crime and improve operational efficiency in high-risk zones.

Data Engineering

Pipeline & Preparation

[Raw Data] → [Cleaning] → [Structured] → [Analysis Ready]

Source	Cleaning Performed	Key Fields Selected
• The dataset was collected from the Los Angeles Open Data Portal (Public Safety Crime Dataset). • It contains approximately 10,000+ records (40+ columns) covering the period 2020–2024.	• Removed entries with missing DR_Number (incomplete crime records). • Cleaned demographic fields: invalid ages (≤ 0) and blank gender/ethnicity set to NA. • Standardized date & time to derive Year, Month, and Hour for trend analysis.	• Time Attributes: Year, Month, Hour of occurrence • Location Details: Area name, Reporting district • Crime Information: Crime description, Crime category • Victim Profile: Age, Gender, Ethnicity • Case Outcome: Investigation / case status

Proper data preparation ensures accurate and unbiased analytical results.

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KPI & Metrics Framework for Data-Driven Policing

Metrics designed to support operational decisions

Problem	Question	KPI	Decision
Increasing crime	Is crime rising yearly?	Yearly Crime Change %	Adjust yearly resource planning
High crime concentration in specific areas	Which areas contribute most to total crime?	Top 5 Areas Contribution %	Deploy targeted hotspot policing & allocate patrol resources
Random patrol timing	When crimes occur most?	Peak Hour Distribution	Schedule patrol shifts
Low case resolution	Are cases being solved?	Closure Rate	Improve investigation workflow

Each KPI directly links analysis to actionable law enforcement strategy.

Operational Insights from Crime Data

Crime is Highly Concentrated in Few Areas

- Top hotspot divisions contribute a large share of total incidents
- Focusing resources on these zones can reduce overall crime faster

2022 Was the Turning Point Year

- Highest crime volume recorded
- Needed stronger intervention due to rising risks



Low Case Resolution Rate

- Unresolved cases exceed closed ones
- Better investigation workflows can improve closures

A Few Crime Types Drive Most Crimes

- Top crime categories dominate total reported cases
- Targeted prevention programs can address the majority of incidents

Crime Peaks During Specific Time Windows

- Higher crime in evening/night hours
- Align patrols with peak times for efficiency

Victim Vulnerability Patterns

- Certain age groups are more affected
- Target awareness programs accordingly

Crime occurrence follows repeatable operational patterns across time, location, and category

Segmentation - Advanced Analysis

How We Segmented the Data

Data Segments

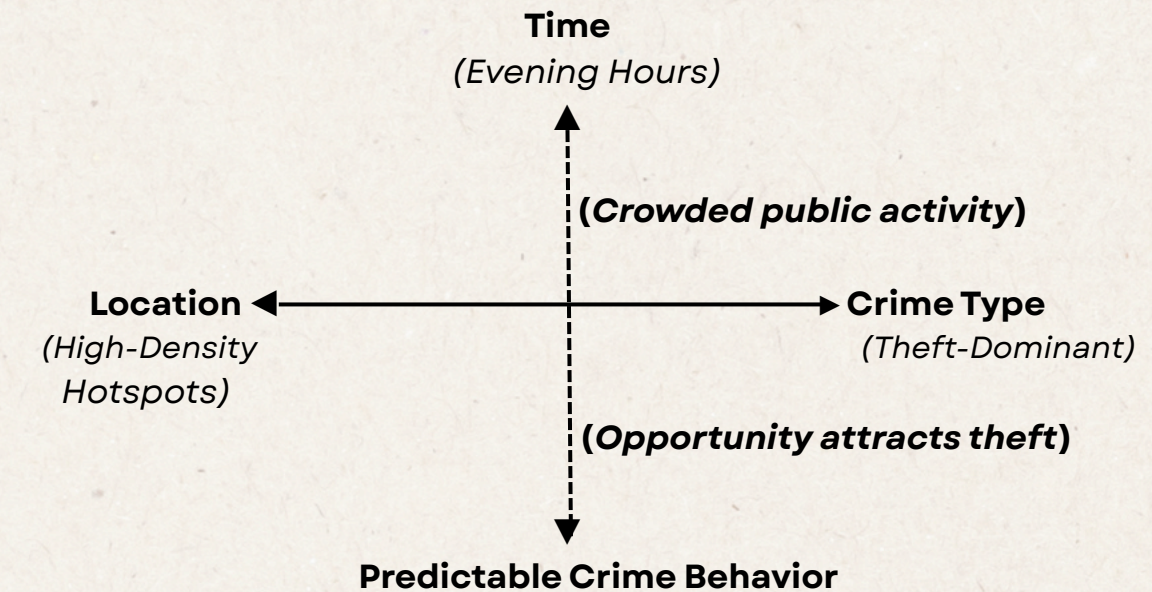
 Year  Area  Hour  Severity  Case Status  Victim Demographics

Key Patterns

- Crime is concentrated in a few hotspot divisions
- Peak hours drive workload, mainly evening/night windows
- Serious crimes dominate, while closure gaps increase operational pressure

Decision Impact

➡ LA policing strategy should shift from uniform coverage to targeted hotspot + peak-hour deployment.



Crime behavior follows predictable opportunity patterns enabling targeted policing decisions



Dashboard Usage Overview

How different users interact with the system

Executive View – City Overview

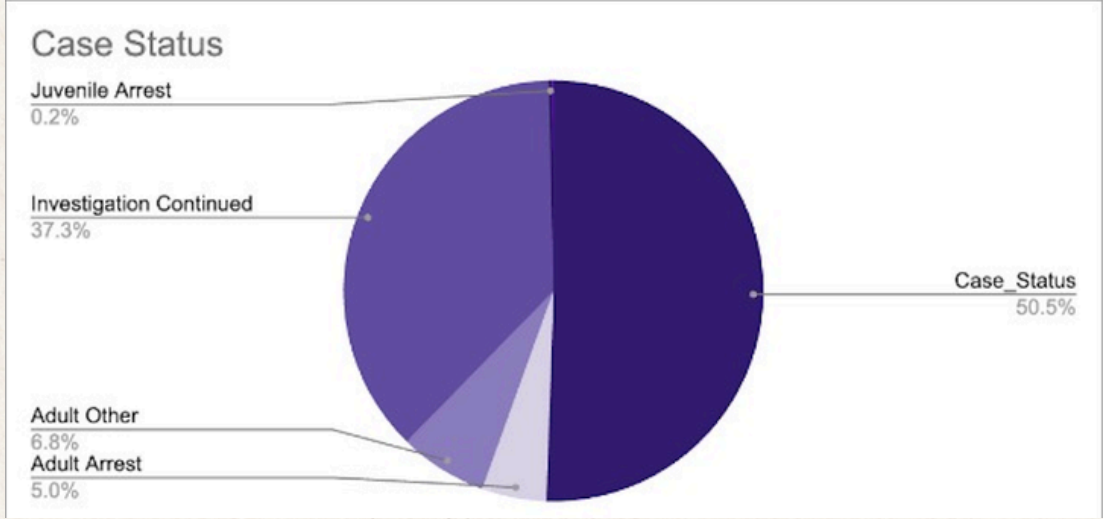
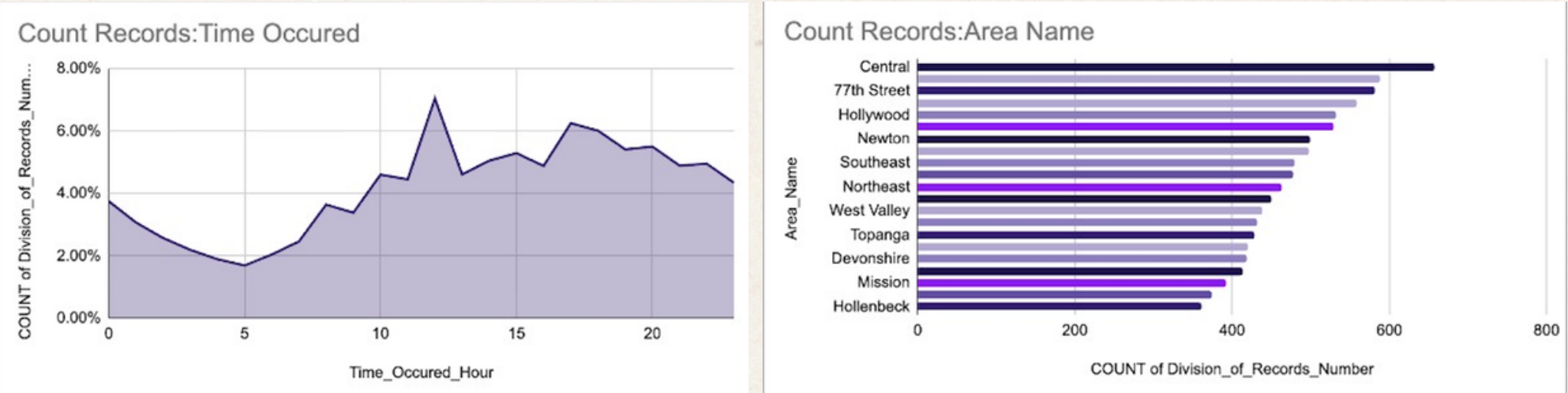
“What is happening overall?”

Crime Data of 2020-2024					
Total Crimes Reported	Highest Crime Year	Lowest Crime Year	Serious Crimes	Less Serious Crimes	Dangerous Zone
9,995	2022	2024	60.30%	39.70%	Central

- Monitor citywide crime trends
- Identify high-risk districts quickly
- Evaluate yearly safety performance
- Support policy and resource planning

Operational View

“Where/when should action be taken?”



Actionable Safety Recommendations

1. Deploy Targeted Evening Patrols

Increase patrol presence during identified peak crime hours to deter recurring incidents.

3. Prioritize High-Frequency Crime Categories

Allocate dedicated investigative resources toward theft-related offenses to improve resolution efficiency.

2. Strengthen Surveillance in High-Risk Areas

Enhance monitoring and security coverage within hotspot districts to reduce concentrated crime activity.

4. Improve Case Resolution Processes

Streamline investigation tracking and reporting workflows to increase closure rates.

5. Implement Preventive Awareness Measures

Conduct public safety outreach in high-risk locations and times to minimize opportunity-driven crimes.



Expected Impact & Stakeholder Value



Recommendation	Operational Impact	Measurable Benefit	Stakeholder Value
Targeted Evening Patrols	Faster response planning	Reduced incident response time	More efficient manpower utilization
Theft Case Prioritization	Faster investigations	Higher closure rate	Improved department performance metrics
Workflow Improvement	Streamlined case handling	Reduced processing delays	Better service reliability

Limitations & Future Scope

What data cannot tell yet...

Data Coverage

- Dataset limited to reported crimes only
- Unreported incidents not captured

Missing Context Factors

- No demographic or socio-economic variables
- External influences (events, holidays, weather) unavailable

Temporal Constraints

- Historical analysis only (2020–2024)
- No real-time predictive capability



Future Improvements

Predictive Modeling

- Forecast crime probability by time and location

Live Dashboard Integration

- Connect with real-time reporting systems

Additional Data Sources

- Incorporate population density, weather, and event data

Decision Automation

- Suggest patrol deployment recommendations automatically

